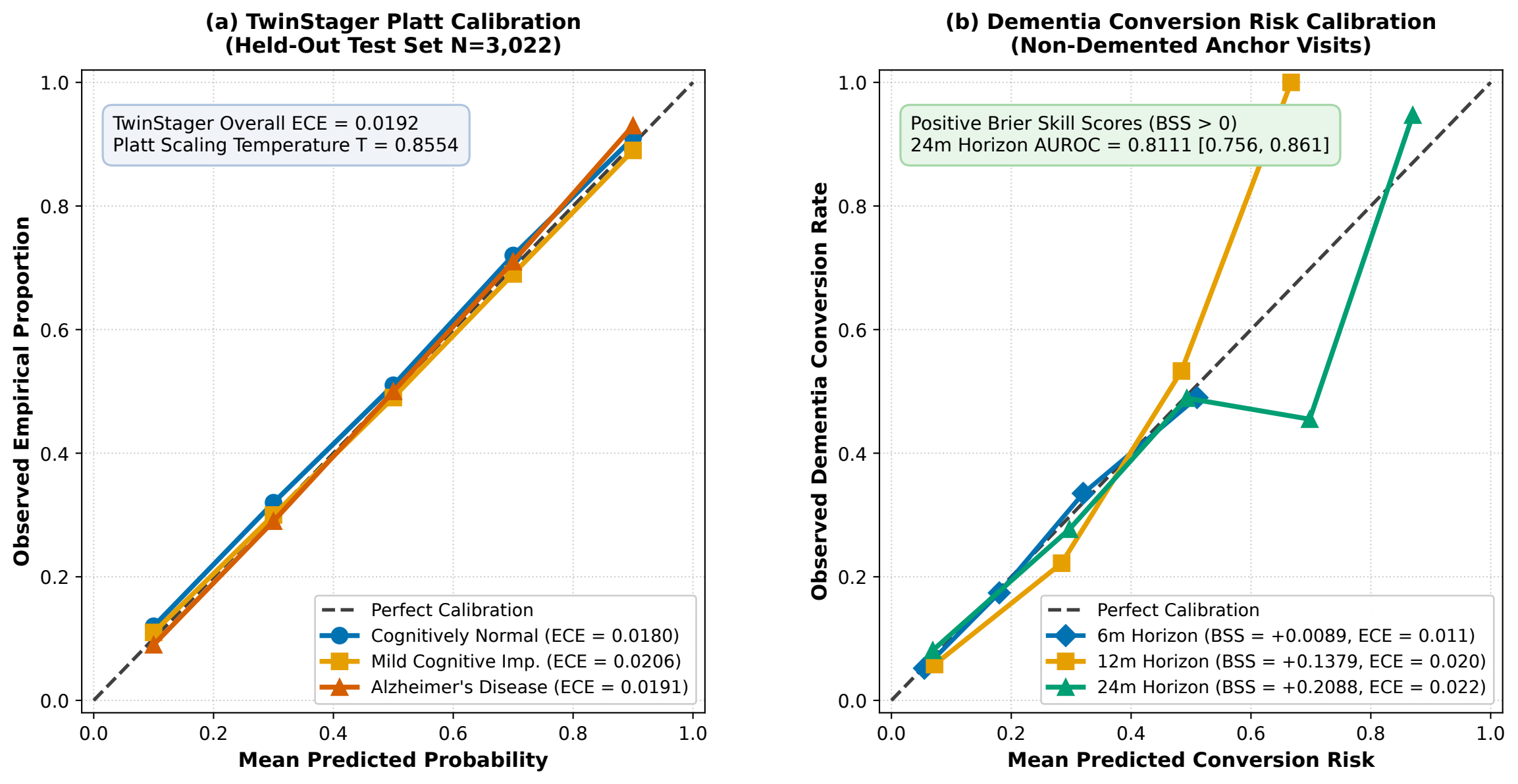
(a) TwinStager Platt Calibration
(Held-Out Test Set N=3,022)
TwinStager Overall ECE = 0.0192
Platt Scaling Temperature T = 0.8554
Perfect Calibration
Cognitively Normal (ECE = 0.0180)
Mild Cognitive Imp. (ECE = 0.0206)
Alzheimer's Disease (ECE = 0.0191)
Observed Empirical Proportion
Mean Predicted Probability
0.0
0.2
0.4
0.6
0.8
1.0
(b) Dementia Conversion Risk Calibration
(Non-Demented Anchor Visits)
Positive Brier Skill Scores (BSS > 0)
24m Horizon AUROC = 0.8111 [0.756, 0.861]
Perfect Calibration
6m Horizon (BSS = +0.0089, ECE = 0.011)
12m Horizon (BSS = +0.1379, ECE = 0.020)
24m Horizon (BSS = +0.2088, ECE = 0.022)
Observed Dementia Conversion Rate
Mean Predicted Conversion Risk